

TRT Pharmacokinetic Analysis

Testosterone Replacement Therapy: Dose-Response,
Cofactor Impact & Protocol Optimisation

Analysis Period:	January 2025 – May 2026
Blood Tests:	6 panels (baseline + 5 on-treatment)
TRT Formulation:	Testosterone Cypionate 200mg/ml, subcutaneous
Current Protocol:	0.26ml × 2/week (72.8 mg active T/week)
Dosing Note:	0.28ml actual (Aug 2025–Feb 2026) due to misread; corrected to 0.26ml
Provider:	Manual (manual.co)
Clinical Framework:	BSSM Guidelines (Hackett et al., 2017; 2023)
Report Generated:	09 May 2026

BSSM On-Treatment Targets

Marker	Target Range	Source
Total Testosterone (trough)	15–30 nmol/L	BSSM 2017/2023
Free Testosterone	0.225–0.500 nmol/L	BSSM / Vermeulen
Haematocrit	< 0.54 (safety ceiling)	BSSM 2017
PSA rise	< 1.4 ng/mL in 12 months	BSSM 2017
Dose titration	10–25% per step, retest 6–12 wks	BSSM 2023

Clinical Question

After 15 months on testosterone replacement therapy with gradual dose escalation (56 → 72.8 mg active T/week), the latest trough total testosterone is 12.7 nmol/L — below the BSSM minimum on-treatment target of 15 nmol/L. This result was obtained at the current prescribed dose (0.26ml × 2/week) and coincides with cessation of a cofactor supplement stack (Boron, Zinc, Magnesium, Selenium, P5P) that had been suppressing SHBG. Note: between Aug 2025 and Feb 2026, actual dosing was 0.28ml due to a misread; this was disclosed to the clinician in Feb 2026 and corrected. This analysis examines whether the current protocol achieves adequate trough levels independently of supplementation, and presents evidence-based options for protocol adjustment.

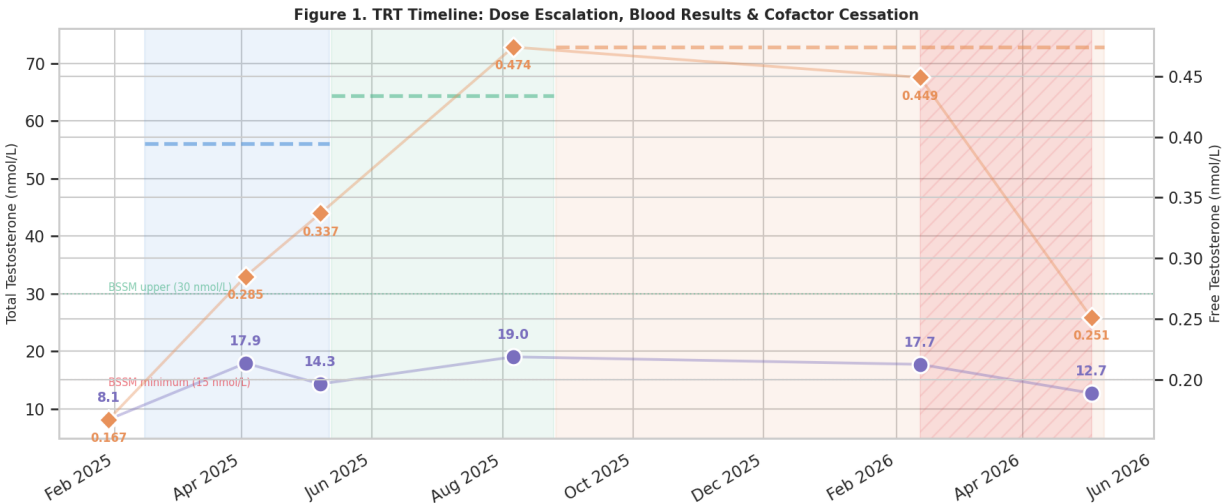
A well-optimised TRT protocol should achieve therapeutic blood testosterone levels independently of supplementation, diet optimisation, or training status. These lifestyle factors should be additive — not compensatory.

1. Blood Test Results Across TRT Phases

Table 1. Blood test results across TRT phases. Red highlights indicate sub-BSSM values. * Active T shows actual dose at time of draw (Feb 2026 = 78.4mg during misread period; May 2026 = 72.8mg after correction).

Date	Phase	Active T mg/wk *	Total T nmol/L	Free T nmol/L	SHBG nmol/L	E2 pmol/L	FAI
2025-01-29	Pre-TRT (Baseline)	0	8.1	0.167	28.6	39	28.4
2025-04-03	Phase 1	56	17.9	0.285	33.8	135	53.0
2025-05-08	Phase 1	56	14.3	0.337	23.2	75	61.6
2025-08-06	Phase 2	64	19.0	0.474	23.2	117	81.9
2026-02-12	Phase 3	78	17.7	0.449	22.5	87	78.7
2026-05-03	Phase 3	73	12.7	0.251	32.1	95	39.6

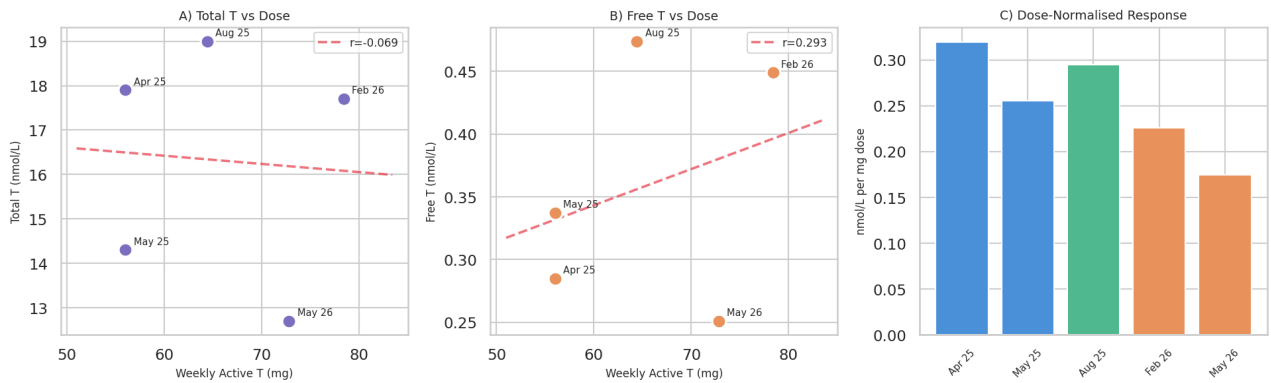
2. TRT Timeline



Dose escalation from 56 to 72.8 mg active T/week over 15 months. Between Aug 2025 and Feb 2026, actual dosing was 78.4 mg/wk due to a misread (corrected Feb 2026). The hatched region indicates cofactor cessation. At the current prescribed dose, the latest result (12.7 nmol/L) falls below the BSSM 15 nmol/L minimum.

3. Dose-Response Analysis

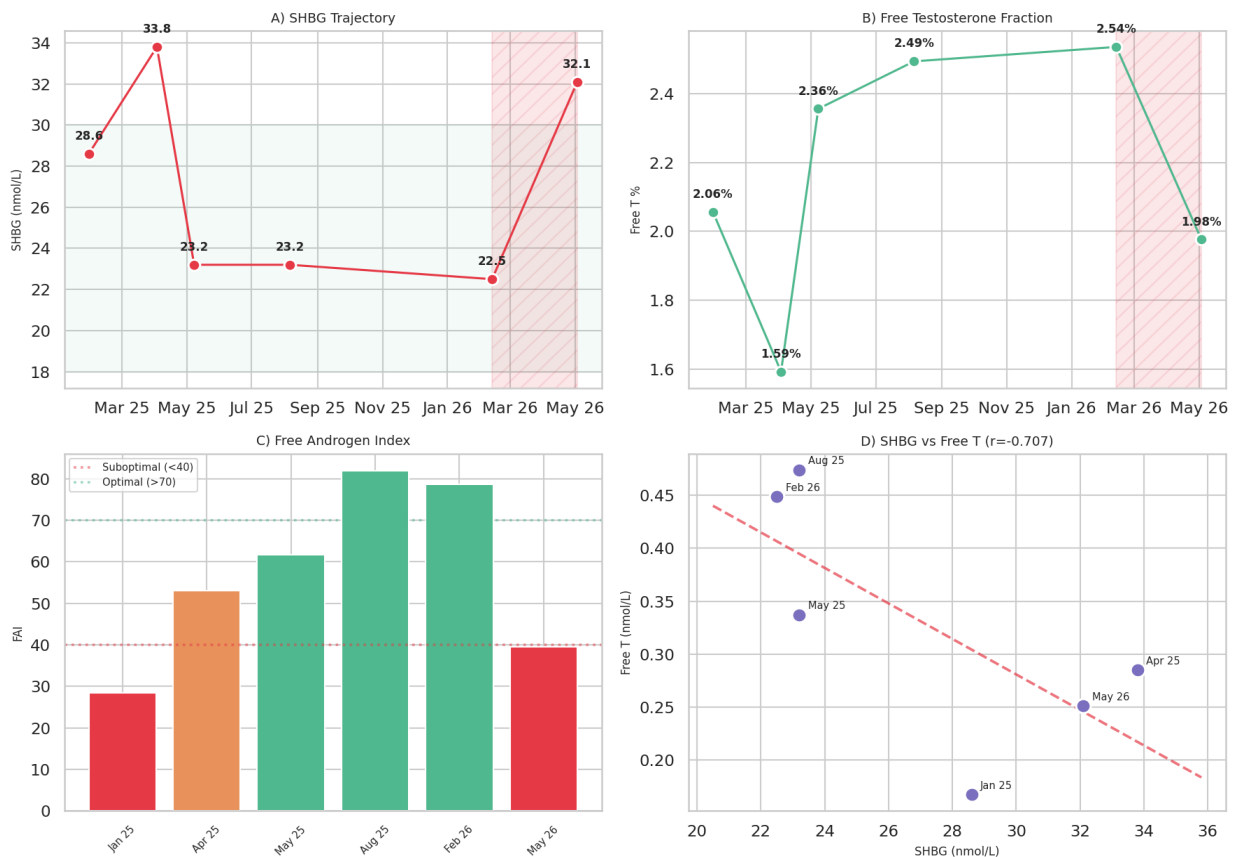
Figure 2. Dose-Response Analysis



Linear regression: Total T = $-0.018 \times \text{dose} + 17.5$ ($r = -0.069$, $R^2 = 0.005$, $p = 0.912$). The paradoxically negative dose-response indicates that confounding variables — primarily SHBG fluctuation driven by cofactor status — dominate over dose as a predictor of serum testosterone.

4. SHBG Dynamics & Free Testosterone

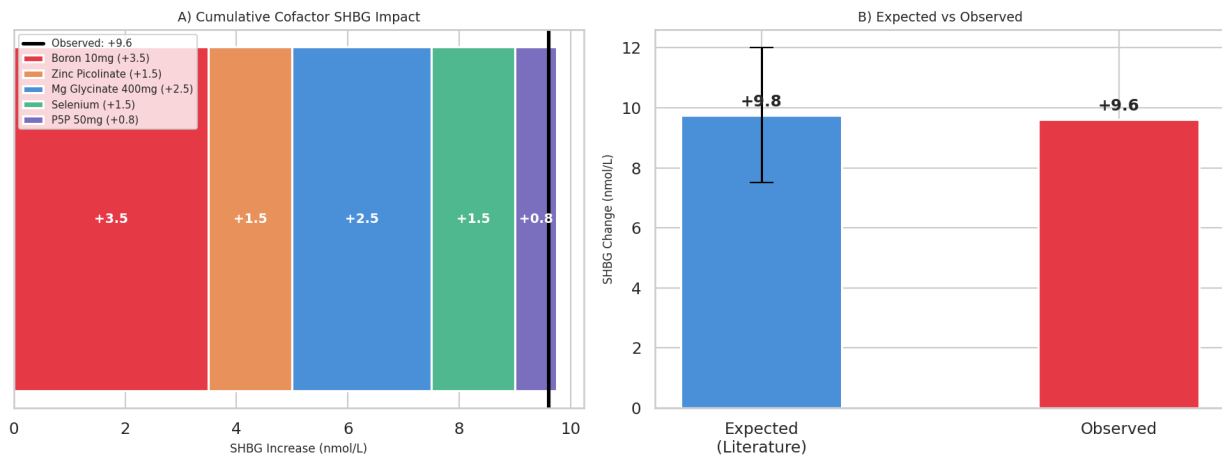
Figure 3. SHBG Dynamics & Free Testosterone Bioavailability



SHBG rose from 22.5 to 32.1 nmol/L (+9.6, +43%) following cofactor cessation. Note: the Feb 2026 draw was at 78.4mg/wk (misread), while the May 2026 draw was at 72.8mg/wk (corrected). The combined effect of cofactor cessation and dose correction drove total T down 28% and free T down 44% ($0.449 \rightarrow 0.251$ nmol/L) — a disproportionate free T decline confirming SHBG-mediated sequestration.

5. Cofactor Cessation Impact

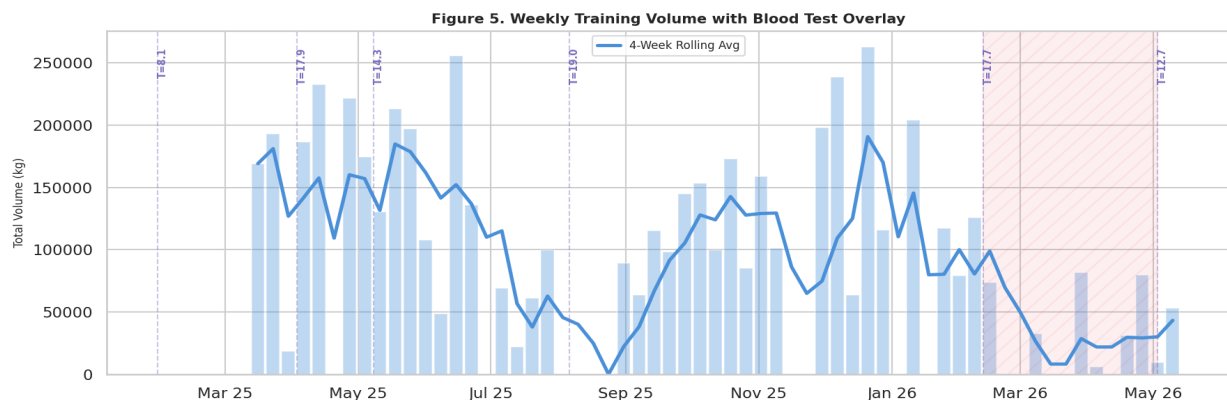
Figure 4. Cofactor Cessation Impact on SHBG



Five supplements were stopped between Feb–May 2026: Boron 10mg, Zinc Picolinate, Magnesium Glycinate 400mg, Selenium, and P5P 50mg. The observed SHBG increase (+9.6 nmol/L) falls within the expected literature range (+7.5–12.5 nmol/L) from cumulative cessation effects (Naghii et al., 2011; Prasad et al., 1996; Excelsmyer et al., 2021).

Key observation: Prior blood results showing adequate levels ($T=17.7\text{--}19.0$ nmol/L) were achieved only with cofactor-suppressed SHBG ($\sim 22\text{--}23$ nmol/L) and, in the case of Feb 2026, a higher actual dose (78.4mg vs prescribed 72.8mg). With natural SHBG (~ 32 nmol/L) and the prescribed dose, testosterone is sub-therapeutic.

6. Training Load

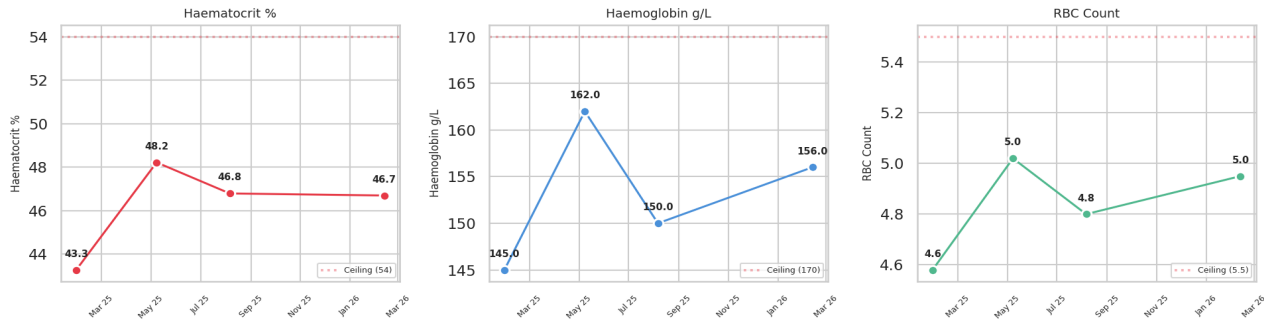


Training volume decreased during the cofactor-off period due to injury. While reduced resistance training can modestly lower SHBG (typically 1–3 nmol/L in eugonadal men; Kraemer & Ratamess, 2005), this represents <30% of the observed +9.6 nmol/L SHBG surge. The dominant driver of the SHBG increase — and consequent free T collapse — is cofactor cessation, not training reduction.

Recovery context: Oura-derived readiness scores and HRV balance remained stable across the analysis period (data not shown), confirming the testosterone decline was not driven by deteriorating recovery or increased physiological stress.

7. Haematological Safety Markers

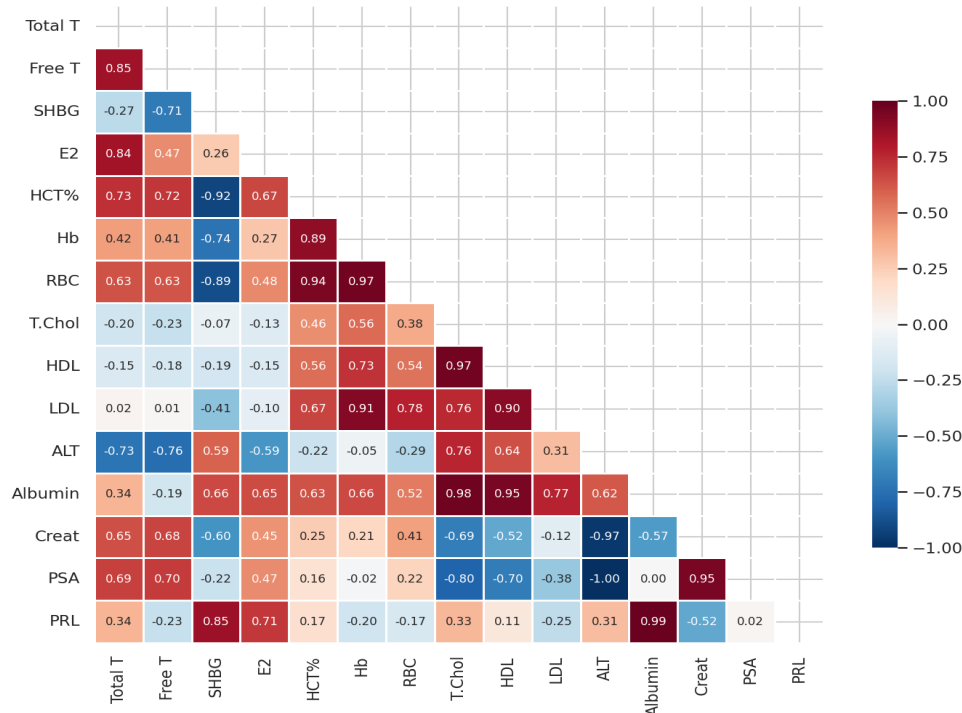
Figure 7. Haematological Safety Markers



All haematological markers remain within safe limits. Current haematocrit is 46.7%, providing **7.3 percentage points of headroom** below the BSSM safety ceiling of 54%. A 25% dose increase typically raises haematocrit by 1–3 percentage points (Bachman et al., 2014), well within the available margin. Even at the upper bound of expected increase, haematocrit would remain below 50%.

8. Blood Marker Correlations

Figure 10. Blood Marker Correlation Matrix



Pearson correlation matrix across available blood markers. Notable relationships include the strong inverse correlation between SHBG and free testosterone, and positive associations between total T and haematological markers (haematocrit, haemoglobin).

9. Injection Frequency Pharmacokinetic Modelling

Testosterone Cypionate has a subcutaneous half-life of approximately 8 days (Kaminetsky et al., 2015). Using exponential decay superposition (Bateman equation), steady-state serum curves were modelled for 2x/week and 3x/week injection frequencies at various doses.

Figure 8. Injection Frequency Pharmacokinetic Modelling

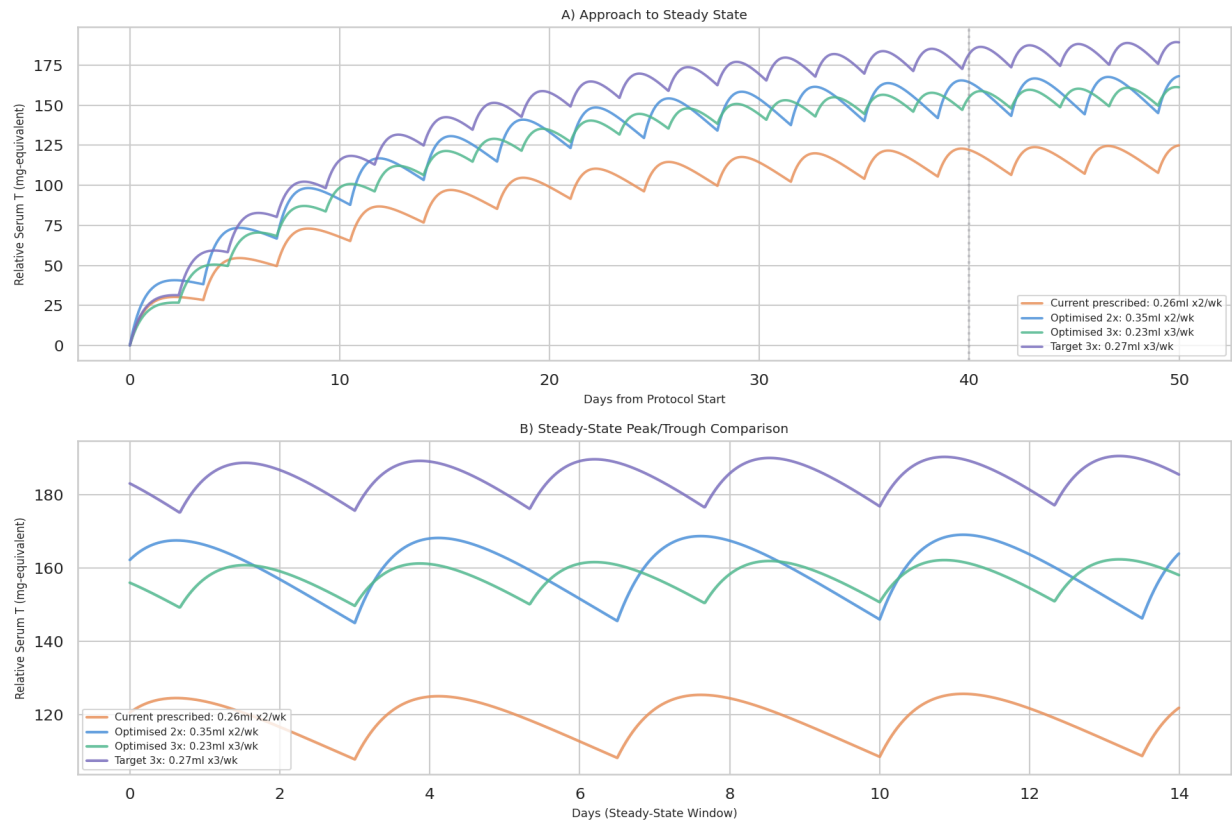


Table 2. PK comparison of injection frequency protocols.

Protocol	Weekly mg Active T	Peak:Trough	Fluctuation
Current prescribed	72.8	1.17:1	15.1%
Optimised 2x	98.0	1.17:1	15.1%
Optimised 3x	96.6	1.09:1	8.3%
Target 3x	113.4	1.09:1	8.3%

Switching from 2x/week to 3x/week reduces peak-trough fluctuation from ~15% to ~8%, producing smoother serum levels. This may reduce oestradiol spikes, improve subcutaneous absorption, and lower haematocrit risk from peak concentrations (Bachman et al., 2014).

10. Dose Optimisation

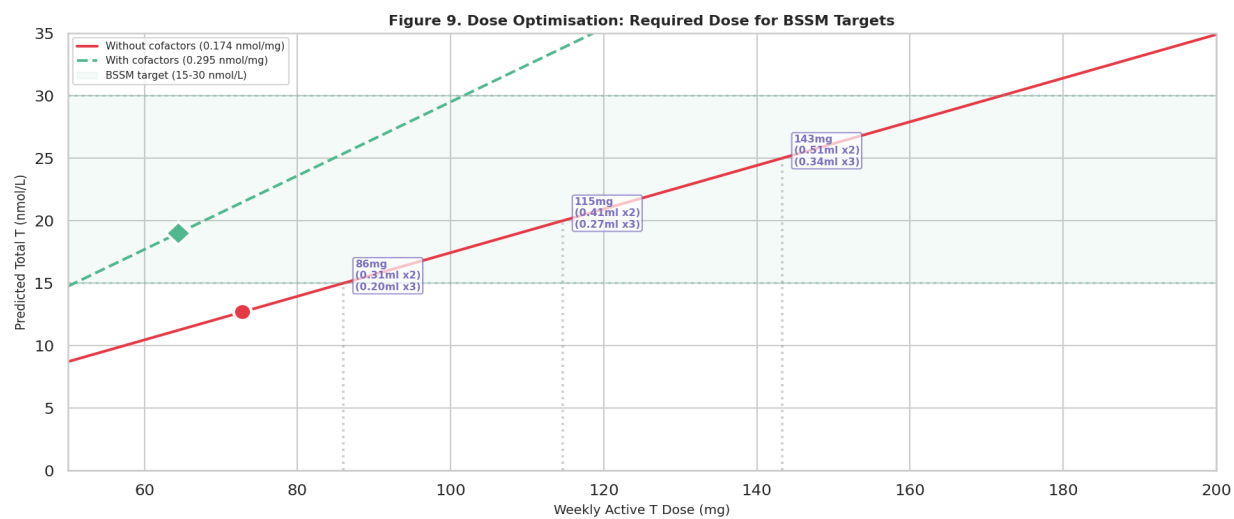


Table 3. Dose-response efficiency comparison.

Scenario	Actual Dose	SHBG	Total T	Ratio
With cofactors (Aug 2025)	64.4 mg/wk	23.2	19.0 nmol/L	0.295 nmol/mg
Without cofactors (May 2026)	72.8 mg/wk	32.1	12.7 nmol/L	0.174 nmol/mg

Table 4. Required dose for BSSM targets without cofactor support.

Target	Active T mg/wk	Cypionate mg/wk	2x/wk (ml/inj)	3x/wk (ml/inj)
15 nmol/L	86	123	0.31	0.20
20 nmol/L	115	164	0.41	0.27
25 nmol/L	143	205	0.51	0.34

11. Discussion Points for Clinician Review

A. Dose Titration

The BSSM practical guide (Hackett et al., 2023) recommends dose increases of 10–25% with retesting at 6–12 weeks when trough levels are below target. The current prescribed dose is 72.8 mg active T/week (0.26ml × 2/week). A 25% increase would yield ~91 mg active T/week (130 mg cypionate, 0.33ml × 2/week or 0.22ml × 3/week), predicting a trough of approximately 15.9 nmol/L — reaching the BSSM minimum target. A second titration step may be needed to reach mid-range targets.

B. Injection Frequency

Switching from 2x/week to 3x/week maintains the same weekly dose while reducing peak-trough fluctuation from ~15% to ~8%. Clinical benefits may include smoother energy and mood, reduced oestradiol conversion at peaks, smaller per-injection volume improving subcutaneous absorption, and lower peak-driven haematocrit risk (Kaminetsky et al., 2015; Bachman et al., 2014). This could be considered as a protocol refinement alongside dose adjustment.

C. Cofactor Management

If cofactors are resumed, they should be viewed as additive to an already-adequate base dose — not required for achieving minimum therapeutic levels. The goal is a protocol that reliably produces BSSM-range trough levels on

its own, with cofactors providing further optimisation into the upper range. This creates a resilient protocol: if supplements are temporarily stopped, levels remain therapeutic rather than crashing to sub-therapeutic as observed in this analysis.

D. Safety Considerations

Haematocrit: Current level (46.7%) provides 7.3 percentage points of headroom below the BSSM 54% ceiling. A 25% dose increase would be expected to raise HCT by 1–3 points, remaining well under 50%.

PSA: PSA has risen 0.79 µg/L over 12.5 months (0.77 → 1.56 µg/L, baseline Jan 2025 to Feb 2026). Annualised velocity: ~0.76 µg/L/year, well below the BSSM referral threshold of 1.4 µg/L in 12 months. Note: PSA was not measured at the May 2026 draw.

Monitoring: A full blood panel is already scheduled for August 2026 (<12 weeks from any protocol change), satisfying the BSSM retest requirement. This will confirm haematocrit, PSA, and testosterone response to the adjusted dose.

12. Recommended Protocol Change

Based on the analysis above, I am requesting the following protocol adjustment:

	Primary Option (Recommended)	Alternative
Dose per injection	0.22ml × 3/week	0.33ml × 2/week
Weekly active T	91 mg	91 mg
Increase from prescribed	25% (BSSM guideline max)	25% (BSSM guideline max)
Frequency change	2x → 3x/week	No change
Peak:Trough ratio	~1.09:1 (smoother)	~1.17:1 (larger swings)
Predicted trough T	~16 nmol/L	~16 nmol/L
Predicted Free T *	~0.40 nmol/L (BSSM range)	~0.40 nmol/L (BSSM range)
Retest interval	August 2026 (planned)	August 2026 (planned)
Trade-offs	Smaller injection volume, flatter serum curve	Larger per-injection volume, higher peak-driven E2/HCT

** Free T estimate assumes SHBG normalises to ~25 nmol/L with cofactor restart (Vermeulen calculation: Total T ~16, SHBG ~25, Albumin ~45 g/L). BSSM Free T range: 0.225–0.500 nmol/L.*

Rationale: A 25% increase is the maximum single-step titration recommended by BSSM (Hackett, 2023). The primary option adds 3x/week frequency to reduce peak-trough fluctuation (–47%), which may improve tolerability and reduce oestradiol spikes. The predicted trough of ~16 nmol/L would reach the BSSM minimum target of 15 nmol/L. A full blood panel is already planned for August 2026 (<12 weeks), providing a built-in safety check within the BSSM-recommended 6–12 week retest window. If August retest shows trough below 18 nmol/L (BSSM lower-mid range), a second 25% titration step would be clinically appropriate, taking weekly active T to ~114 mg and projected trough to ~20 nmol/L (BSSM mid-range).

If cofactors are resumed alongside a dose increase, levels should reach mid-range (~20 nmol/L) rather than hovering at the minimum — building the resilient protocol where supplements are additive, not compensatory.

References

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